DATA SCIENCE & BUSINESS ANALYTICS

Markdown Program for Close to Expiration Products

EDIT. DISRUPTIVE DIGITAL EDUCATION

JULHO 2026
PORTO

Optimizing the daily markdown process enables a better balance between minimizing losses and protecting gross margin

The Perishables Challenge

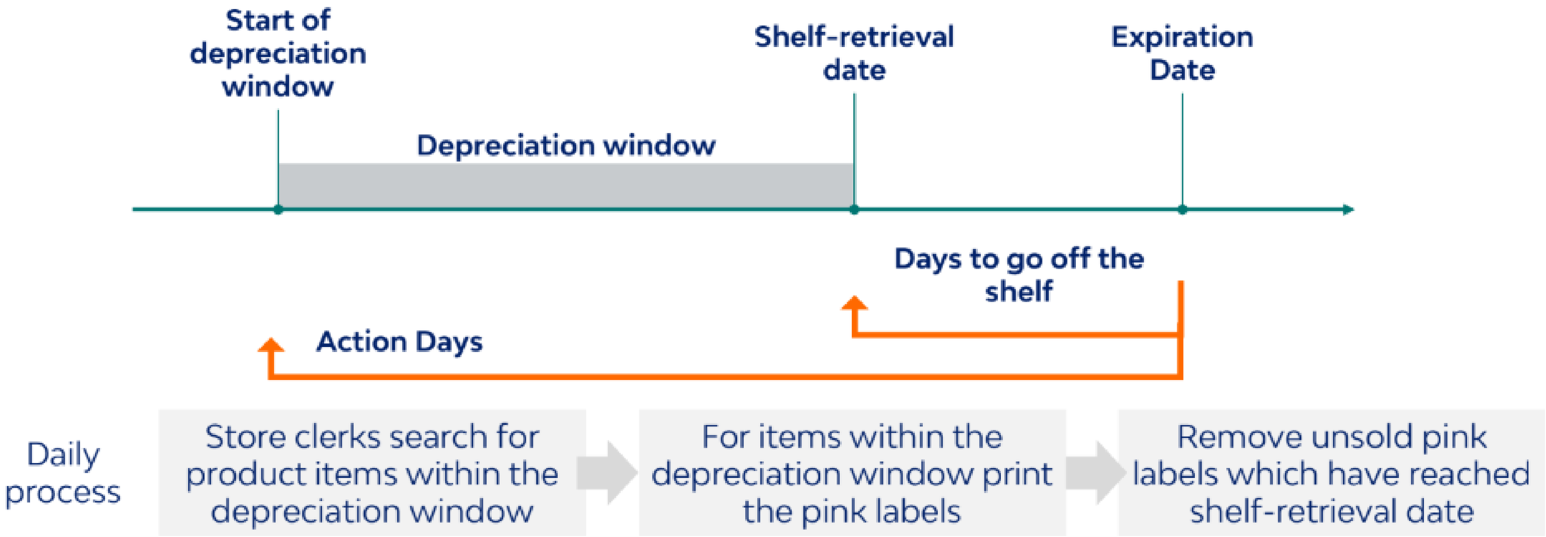
Grocery retail faces a constant struggle with products approaching expiration.

The Discount Dilemma

Discounting too late leads to waste; discounting too early unnecessarily erodes gross margin.

Objective

Predict the optimal timing and right markdown depth to maximize conversion rates.



Analysis of 150k+ historical records uncovered missing values, inconsistencies, and the need for data preparation

DATASET	DATA_STORE	DATA_LABELS
TABLE TYPE	Dimension	Facts
RELATIONSHIP	1	Many (via id_store)
DESCRIPTION	Stores	Labelling and Sales
GRANULARITY	1 row per store	1 row per labelled SKU
COLUMNS	4	15
TARGET	-	Sold
ROWS	342	150054

COLUNAS	INCONSISTENCIES
TYPE	Missing Values
SELLING_SQUARE_FT	Missing Values

COLUNAS	INCONSISTENCIES
NEW_PVP (DISCOUNT)	Concatenated Values & Mixed Formatting
BRAND	Mixed Formatting
EXPIRING_DATE	Date Format
LABELLING_DATE	
SELL_DATE	
OLD_PVP	Outliers
NEW_PVP	

Data inconsistencies were addressed to ensure reliable downstream analysis

DATA_STORES

DIMENSION

Global vs. Regional Imputation

Missing values in `type` and `selling_square_ft`. The proximity between mean and median justified imputing via median for standard cases.

Geographic Exception (Store 194 - Vila Real)

Physical area influenced by population density and local cost per m^2. The global median was rejected to avoid distortions for Store 194.

Implemented Solution

Area imputation based on the median of stores in the same district (Vila Real) and type via regional mode, ensuring market consistency in the dashboard.

DATA_LABELS

FACTS

Feature Extraction

Separation of the text field `new_pvp (discount)` into two independent numerical columns: `new_pvp` and `discount`.

Attribute Normalization

Removal of the redundant "Marca" prefix across all rows in the `brand` column.

Percentage Standardization

Conversion of the `discount` field from text format to decimal percentage (e.g., from 30% to 0.30).

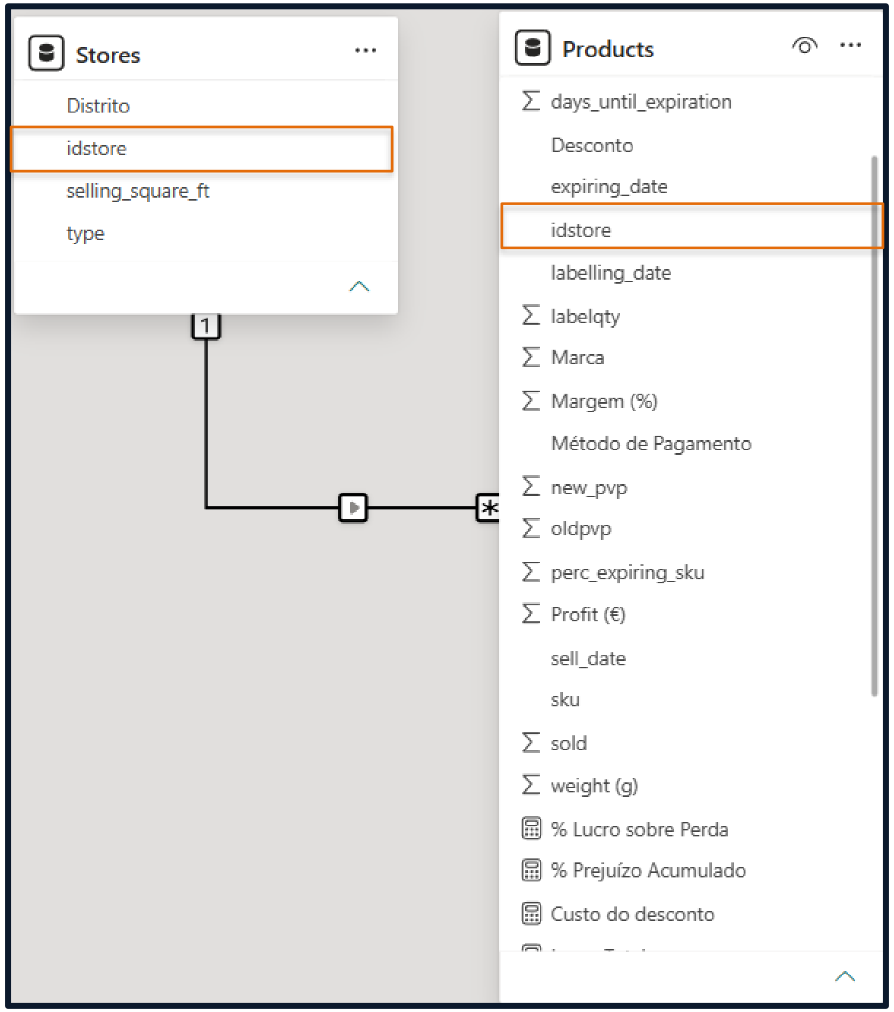
Data Types

Conversion to `datetime` for the `expiring_date`, `labelling_date` and `sell_date` columns.

Business Insights and Visualization

Dashboard

A robust semantic model and business metrics power the dashboard



CREATED METRICS	DESCRIPTION
T PRODUCTS SOLD	Sum of sold products
T PRODUCTS UNSOLD	Sum of unsold products
CONVERSION RATE	% of labelled SKUs sold relative to total SKUs
WASTE RATE	% of labelled SKUs unsold relative to total SKUs
TOTAL REVENUE	Total revenue (€) generated from the sale of labelled SKUs
LOST REVENUE	Total potential sales revenue (€) lost due to unsold SKUs
TOTAL PROFIT	Net profit margin (€) generated exclusively from SKUs sold at discount
LOSS	Total cost (€) associated with unsold SKUs
DISCOUNT COST	Total value (€) given to customers in discounts
% AVOIDED LOSS	% of total potential revenue successfully recovered by selling SKUs at a discount

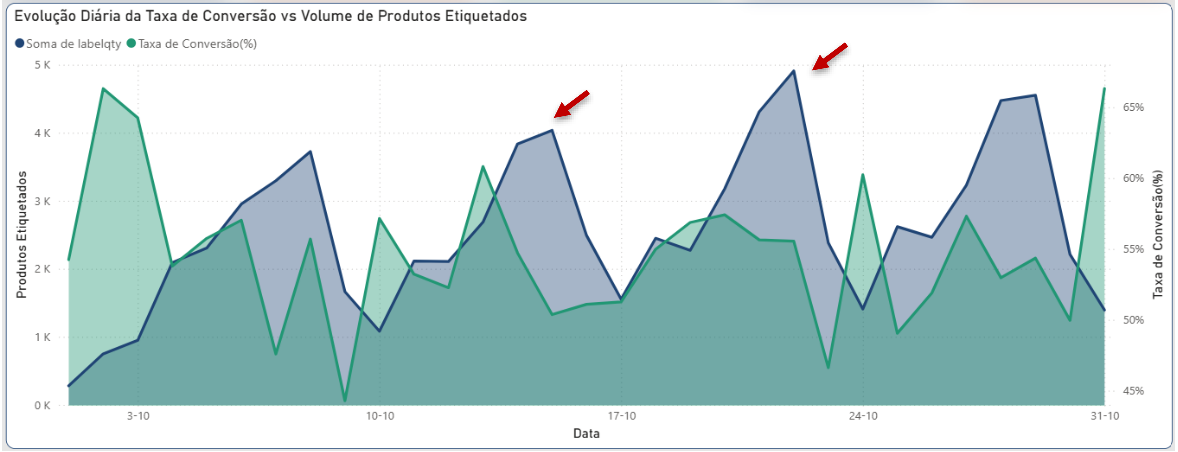
Markdown effectiveness depends on where and when discounts are applied

85,21 K€ TOTAL REVENUE	79,93 K€ LOST REVENUE	13,48 K€ TOTAL PROFIT	53,7% CONVERSION RATE	15,8% AVERAGE MARGIN

What % of labelled SKUs are wasted?

46,3%

Labelling more does not mean selling more

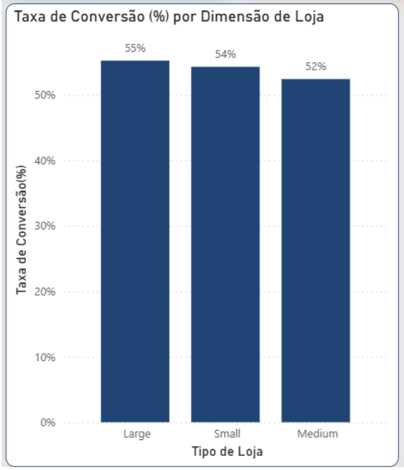


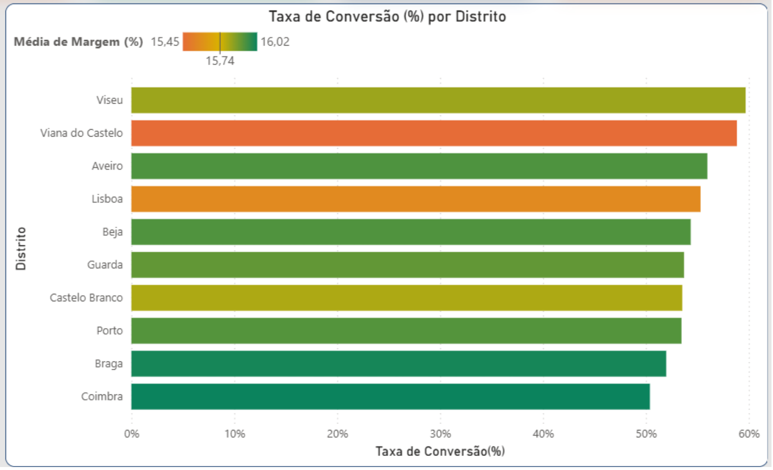
What impacts conversion rate the most?

3 p.p.
Variation across Store Types

10 p.p
Variation across Districts

Location appears to have a greater impact on conversion rate than physical store size





Markdown performance depends on timing and targeting, and not on deeper discounts

0,9
AVERAGE DAYS TO SELL

37,4 K€
DISCOUNT COST

Increasing the discount does not lead to higher Conversion Rates

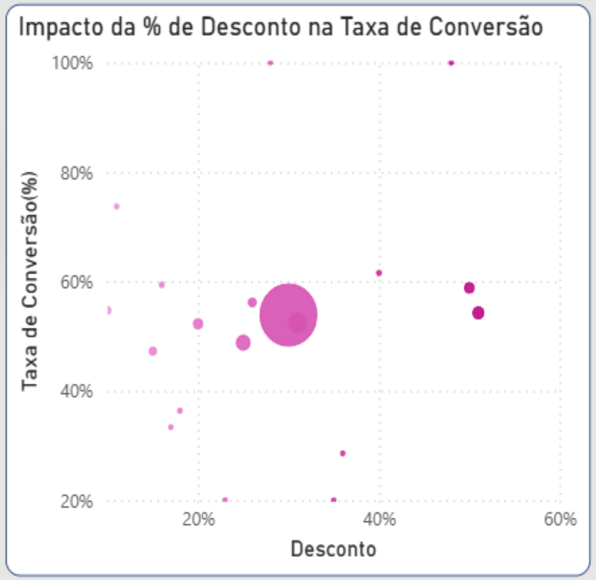
30,2 %
AVERAGE DISCOUNT

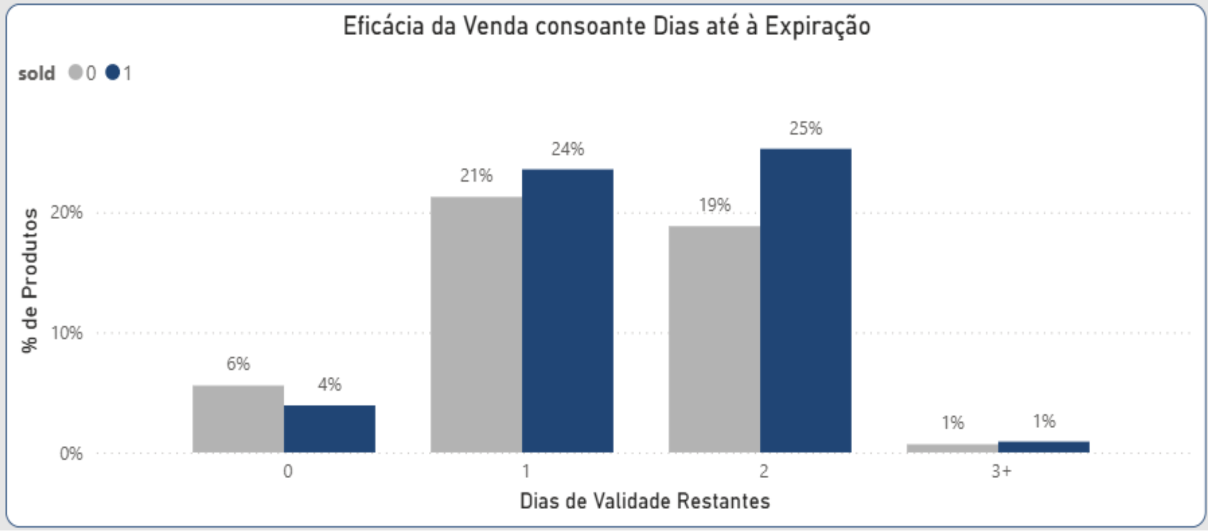
Which brands do the SKUs with the highest waste volume belong to?

Marcas 2 e 15

When should the label be applied to maximize sales?

Up to 48h
BEFORE EXPIRATION





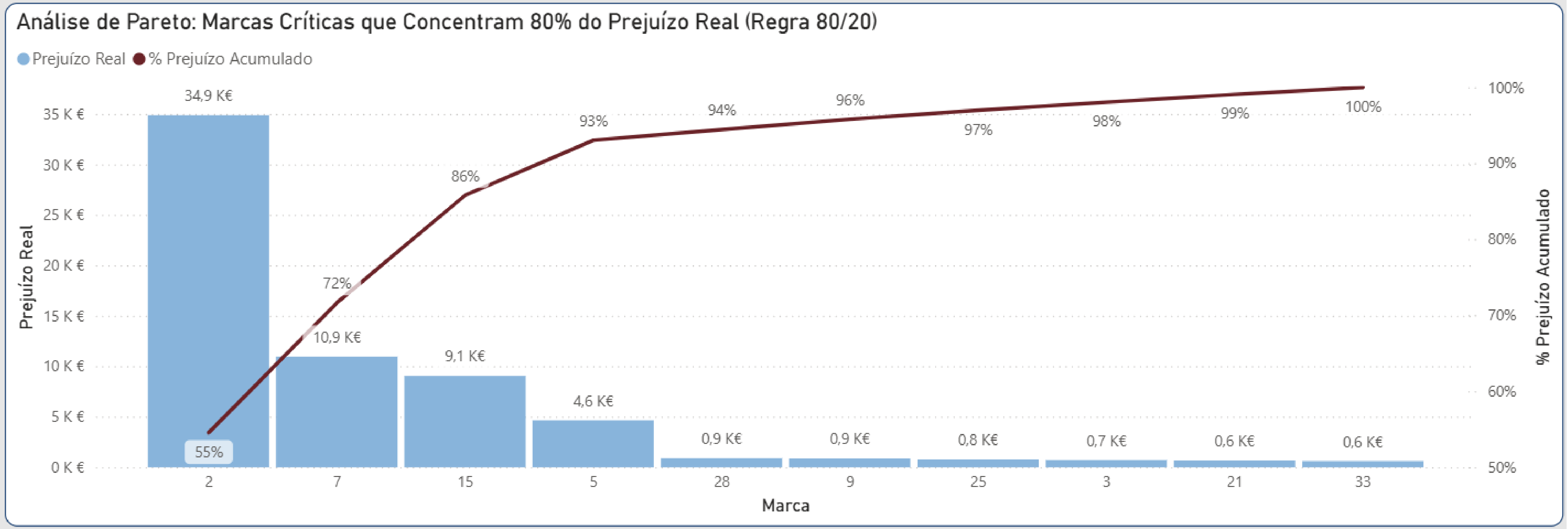
Focusing on three brands delivers the greatest opportunity to reduce losses

Just 3 brands account for over 80% of the loss.

It is not worth spending operational effort trying to adjust the entire catalog. Markdown profitability is resolved by intervening in the supply cycle of the 3 main brands.

Markdown recover 51,6% of potential loss.

Markdowns already recover half the value, but 48.4% of the value at risk is still being lost.



Modeling

Segmentation

Data preparation and validation to ensure the robustness of the store segmentation

DATA_LABELS

FACTS

Feature Engineering:

Created the metric “days_until_expiration”: days remaining until the product expires from the moment it was labelled.

Duplicate Check

Kept because the business confirmed that the exact same sale can occur simultaneously.

Outlier Removal

Dropped datapoints where old_pvp, new_pvp, and discount were errors (e.g., €999 as old_pvp).

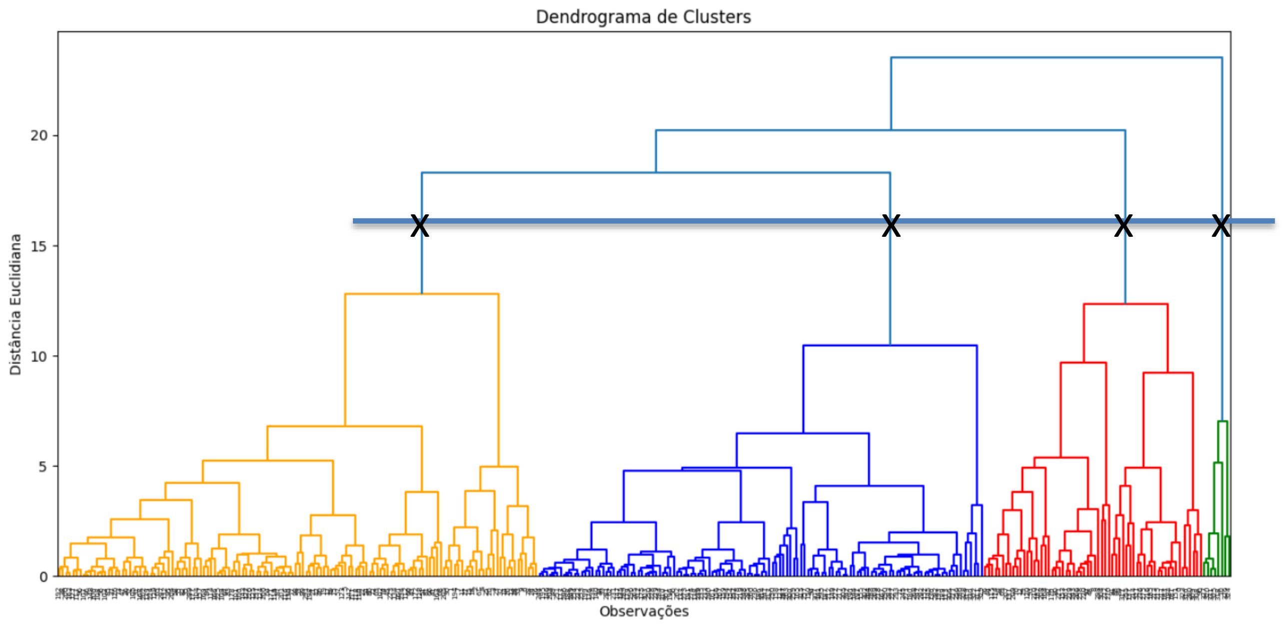
Inconsistency Removal

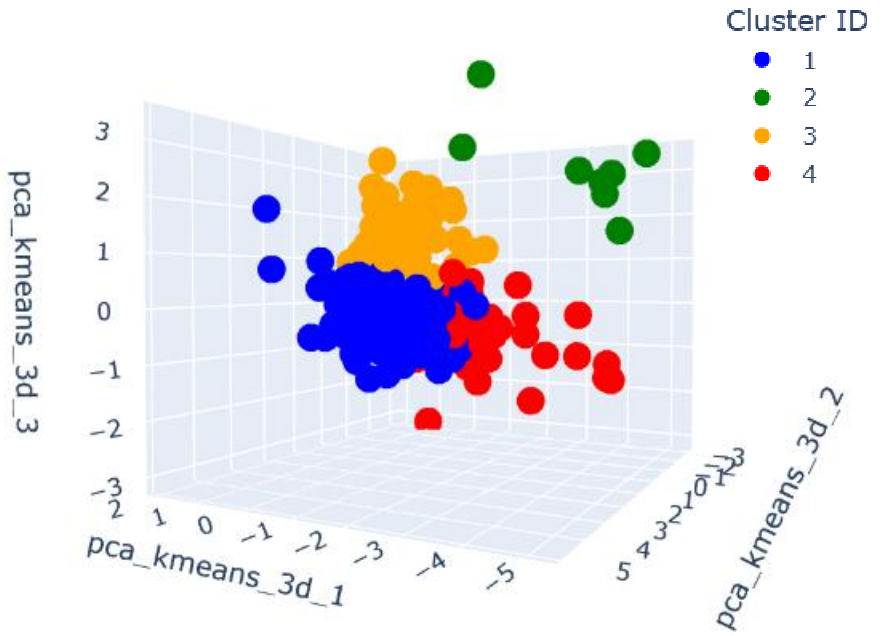
Dropped datapoints where the labelling date is later than the sale date.

Four key operating profiles define the network, distinguished mainly by size and discount policy

How can we cluster our stores based on their **performance**?

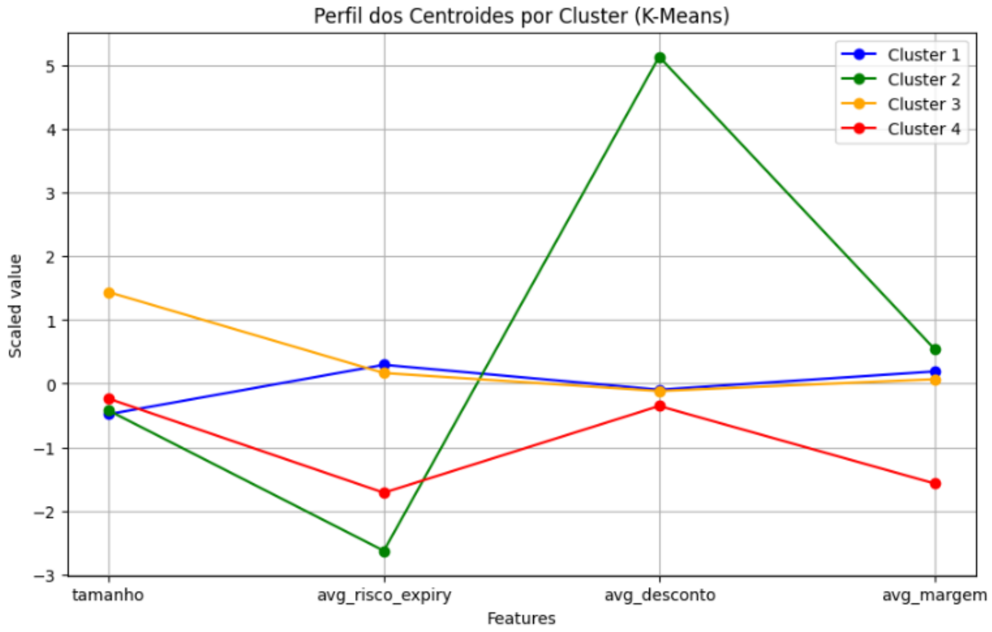
Store Size	Average Expiration Risk	Average Discount	Average Margin





Despite homogeneous profitability, the segmentation reveals major discounting anomalies and differences in scale

How can we cluster our stores based on their **size**, **expiration risk profile**, **applied discounts**, and **margin**?



ID	Profile	Stores	Size (sq ft)	Expiration Risk	Discount	Margin
1	Core Standard Operations	213	Medium	Medium	Medium	Medium
2	Premature Discounters	8	Medium	**Very Low**	**Very High**	Medium
3	High-Volume Powerhouses	79	**Very High**	Medium	Medium	Medium
4	Low Margin Operations	32	Medium	Low	Medium	Low

- A **Store Network is homogeneous in margin.**
- The resulting clusters primarily identified **discount anomalies** and **size differences**:
  - **Cluster 2** revealed 8 stores applying higher discounts far too early (~49% discount e ~ 7 days in advance).
  - **Cluster 3** separated larger stores rather than stores with a distinct labelling strategy.

Modeling

Classification

The initial feature set combined business knowledge with engineered variables

BUSINESS FEATURES

NEW_PVP

MARGIN

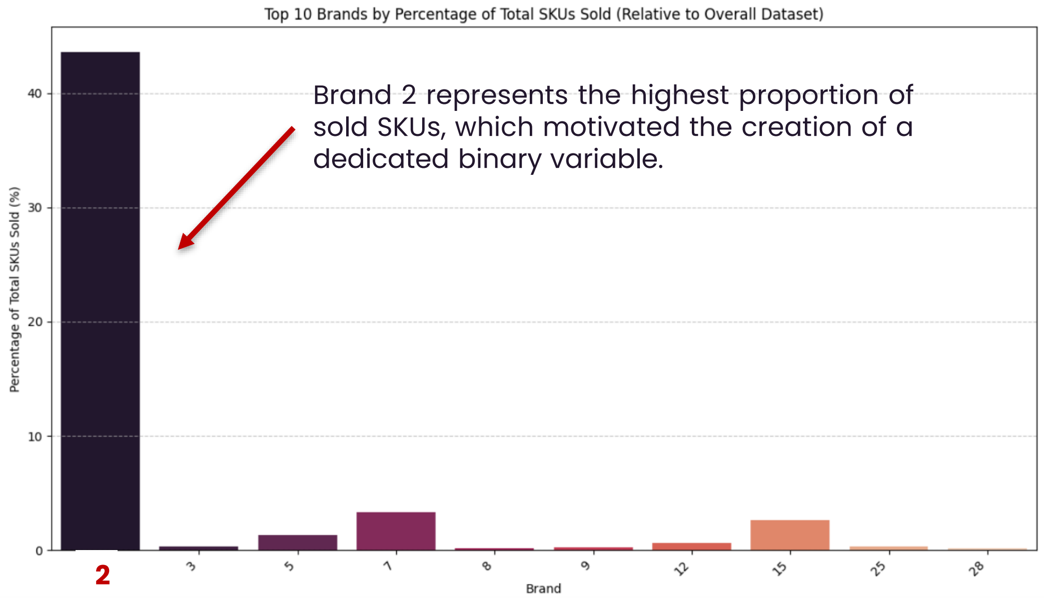
PROFIT

DISCOUNT

SELLING_SQUARE_FT

NEW ENGINEERED FEATURES

- **DAYS_UNTIL_EXPIRATION:** days remaining until the product expires from the moment it was labelled.
- **IS_WEEKEND_LABELLING:** variable that identifies whether labelling occured over the weekend.
- **IS_BRAND_2:** binary variable through *one-hot encoding* to indicate whether the producto belongs to Brand 2.



Model Selection

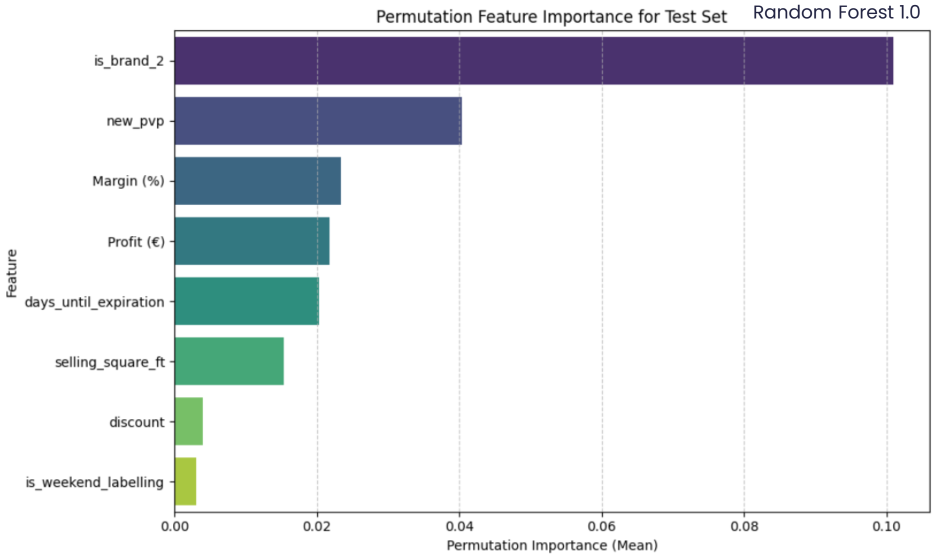
ITERATION	RANDOM FOREST 1.0	RANDOM FOREST 2.0	RANDOM FOREST 3.0	XGBOOST
CHANGE	-	FEATURE SELECTION + HYPER PARAMETER TUNNING	**HYPER PARAMETER TUNNING**	FEATURE SELECTION
FEATURES	DISCOUNT MARGIN PROFIT NEW_PVP DAYS_UNTIL_EXPIRATION IS_WEEKEND_LABELLING SELLING_SQUARE_FT IS_BRAND_2	MARGIN PROFIT NEW_PVP DAYS_UNTIL_EXPIRATION	DISCOUNT MARGIN PROFIT NEW_PVP DAYS_UNTIL_EXPIRATION IS_WEEKEND_LABELLING SELLING_SQUARE_FT IS_BRAND_2	MARGIN PROFIT NEW_PVP DAYS_UNTIL_EXPIRATION
ACCURACY	0,67	0,66	**0,69**	0,65
RECALL	0,66	0,65	**0,69**	0,65
PRECISION	0,67	0,65	**0,69**	0,65
F1-SCORE	0,66	0,65	**0,69**	0,65
ROC-AUC	0,718	0,704	**0,758**	0,704

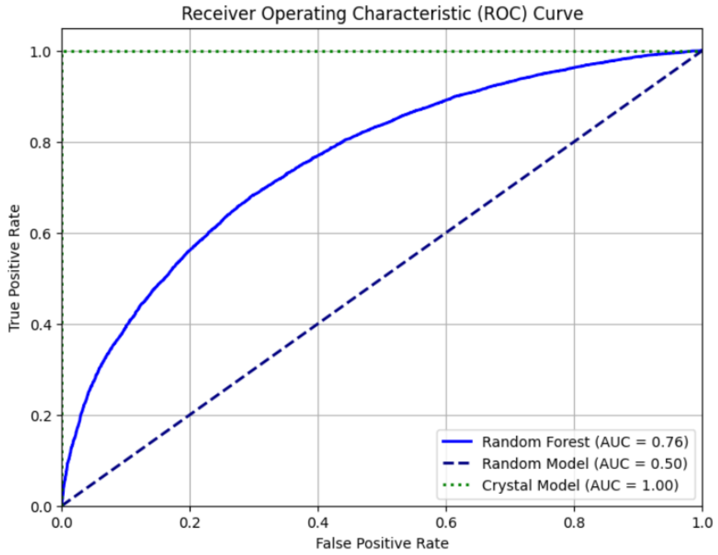
Random Forest 3.0 accurately predicts sales without overfitting

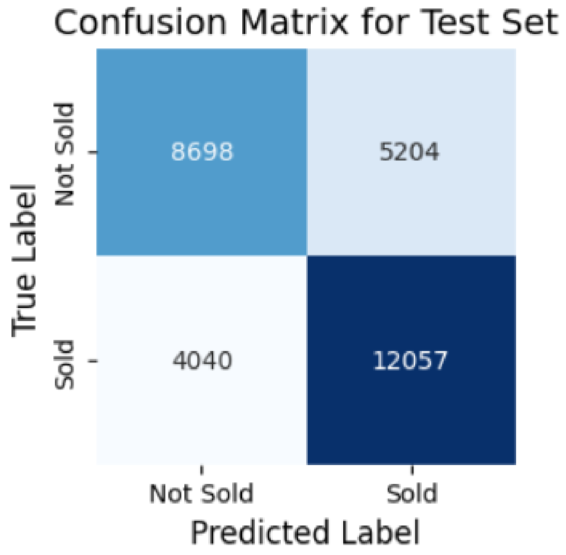
	FEATURES	ACCURACY	RECALL	PRECISION	F1-SCORE	ROC-AUC
RANDOM FOREST	DISCOUNT MARGIN PROFIT NEW_PVP DAYS_UNTIL_EXPIRATION IS_WEEKEND_LABELLING SELLING_SQUARE_FT IS_BRAND_2	0,69	0,69	0,69	0,69	0,758

ROBUSTNESS

PREDICTIVE PERFOMANCE







Business Recommendations

Key operational levers to protect margin and enhance store efficiency

SUPPLY CHAIN & PROCUREMENT — **>80% of Losses**

Recalculate Supply for the Top 3 Critical Brands

Adjust purchasing volumes and replenishment cycles for **Brands 2, 7, and 15**. Avoid overstocking at the source rather than trying to manage discounts across the entire catalog.

MARKDOWN TIMING — **Até 48h Antes**

Standardize Discount Window (24h-48h Window)

Apply discount labels strictly **48h before expiration**. Sales right at expiration (0 days) fail, while deeper discounts (>30%) do not generate higher conversion.

EFFICIENT LABELLING — **Reduce 46,3% Waste**

Replace Mass Labelling with Selective Labelling

Eliminate the rule of labelling all near-expiration stock ("labelling more does not mean selling more"). Implement a **predictive labelling model** focused only on SKUs with a proven sell-through history.

REGIONAL ALOCATION — **10 p.p. de Impacto**

Reroute Stock to High-Conversion Districts

Prioritize shipping markdown items to stores in **high-conversion districts** (e.g., Viseu, Viana do Castelo), regardless of physical store size (only 3 p.p. variation).

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Thank you!
Questions?

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